Program overview

08-Dec-2022 11:31

Year	**2022/2023**
Organization	**Technology, Policy and Management**
Education	**Bachelor Systems Engineering, Policy Analysis & M**

Code	Omschrijving	ECTS	p1 p2 p3 p4 p5
Bachelor TB 2022			
1e jaar TB 2022	**1e jaar BSc TB 2022**		
TB111C	Problem Analysis	5	
TB112C	Systems Modelling 1: Modelling Principles	5	
TB113B	Systems Modelling 2: Complex Systems	5	
TB121B	Governance and Law 1	5	
TB122B	Micro and Market Economy	5	
TB131B	Differential Equations and Linear Algebra	5	
TB132B	Multivariable Analysis and Linear Algebra	5	
TB133D	Introduction to Programming with Python	5	
TB134A	Statistics and Data Analysis	5	
TB135C	Operational Research	5	
Technologiespecialisatie 1	**Technology Specialisation 1**		
TB141EB	Introduction to Energy and Industry Systems	5	
TB141IC	ICT System Engineering and Rapid Prototyping	5	
TB141TB	Transport System Analysis	5	
Technologiespecialisatie 2	**Technology Specialisation 2**		
TB142EA	Analysis of Energy Systems	5	
TB142IC	Data Science in Technische Bestuurskunde	5	
TB142TA	Logistics 1	5	
2e jaar BSc TB 2022	**2nd year BSc TB 2022**		
TB211A	Analysis of Multi-Actor Systems	5	
TB212B	Ethics and Safety	5	
TB221B	Economy of Infrastructures	5	
TB222A	Governance and Law 2	5	
TB223C	Organisation and Management	5	
TB231C	System modelling 3: simulation methods	5	
TB232B	Research and Data-analysis	5	
TB233B	System modelling 4: project simulation methods	5	
TB234B	Multivariate Data Analysis	5	
Technologiespecialisatie 3	**Technology Specialisation 3**		
TB241EA	Transport Phenomena	5	
TB241IB	Object Oriented Programming	5	
TB241TB	Logistics 2	5	
Technologiespecialisatie 4	**Technology Specialisation 4**		
TB242EB	Processes in the Energy Sector	5	
TB242IB	Interconnected Engineering Systems	5	
TB242TB	Improving the Transport System	5	
Technologiespecialisatie 5	**Technology Specialisation 5**		
CTB1420-17	Transport en Planning	5	
TB243EA	Analysis of Industrial Systems	5	
TB243IB	I&C Systems: Risk & Control	5	
3e jaar BSc TB 2022	**3rd year BSc TB 2022**		
Verplichte vakken 3e jaar 2022			
TB323A	Governance van sociotechnische systemen	5	
TB351D	Bachelor Final Project	15	
Governance specialisatie (E&I / I&C / T&L)	**Governance specialisation (E&I / I&C / T&L)**		
TB321EB	Governance specialisation (E and I)	5	
TB321IB	Governance specialisation (I and C)	5	
TB321TB	Governance specialisation (T and L)	5	
Technologiespecialisatie 6	**Technology Specialisation 6**		
TB341EC	Performance analysis in Energy and Industry	5	
TB341IC	I and C Risk and Control	5	
TB341TC	Quantitative Models for Transport	5	
Keuze minor (zie minor lijst) 2021	**Minor Elective (see minor list) 2021**		

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Bachelor TB 2022	
Program Title	Technische Bestuurskunde
Director of Education	J.A. Annema (Jan Anne)
Prerequisites	
Introduction 1	https://www.tudelft.nl/studenten/faculteiten/tbm-studentenportal/onderwijs/bachelor/welkom-nieuwe-bachelor-studenten

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1e jaar TB 2022	
Director of Studies	Dr. J.A. Annema

TB111C	Problem Analysis	5
Module Manager	Dr. L.J. Kortmann	
Instructor	M.B. van der Meulen	
Co-responsible for assignments	Dr.ir. C. van Daalen	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	

TB112C	Systems Modelling 1: Modelling Principles	5
Module Manager	Dr. P.W.G. Bots	
Instructor	Dr.ir. I. Bouwmans	
Instructor	Dr. P.W.G. Bots	
Responsible for assignments	Dr. P.W.G. Bots	
Co-responsible for assignments	Dr.ir. I. Bouwmans	
Contact Hours / Week x/x/x/x	0/2/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

TB113B	Systems Modelling 2: Complex Systems	5
Module Manager	Prof.dr. M.E. Warnier	
Instructor	Ö. Okur	
Co-responsible for assignments	Prof.dr.ir. J.H. Kwakkel	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	Dit vak verzorgt een introductie in het modelleren en simuleren van complexe socio-technische systemen. De volgende onderwerpen komen aan bod in het vak: complexiteit en socio-technische systemen, fysieke modellen vergeleken met socio-technische modellen, procesmodellen, simulatiemodellen, de modelleer cyclus, het concept agent en agent interactie, emergentie, wachtrijen, stockflows.	
Study Goals	Na succesvolle afronding van de module: - kan de student uitleggen wat een complex systeem is en hoe dit zich dit verhoud tot andere systemen - is de student in staat om de informatie flow van een complex systeem te modeleren en beschrijven mbv een proces diagram in IDEF0/BPMN - is de student in staat om de modelcyclus die nodig is voor het maken van een simulatiemodel te doorlopen (conceptualisatie, specificatie, implentatie, verificatie, validatie, experimenteren enz.); - kan de student van de belangrijkste modelleerconcepten (toestand, regels, invoer, uitvoer, visualisatie enz.) uitleggen hoe die gebruikt worden in basis simulatiemodellen, en kan de student python als simulatiegereedschap efficiënt en effectief gebruiken - is de student in staat een niet te complex model foutloos en binnen beperkte tijd in python te coderen - is de student in staat om voor een simulatie model een experiment plan op stellen dat systematisch het gedrag van het model exploreert, en dat gebruikt kan worden voor het beantwoorden van specifieke onderzoeksvragen	
Education Method	Colleges en practica. Voor de practica wordt Python gebruikt. Het practica bereid voor op de stof van de modelleerlijn in het tweede jaar (wachtrijen, stockflows, emergentie).	
Assessment	Tentamen (70%) en groepsopdracht (30%). Voor beiden onderdelen moet een score >5 worden gehaald. Daarnaast moet de skills module Information Literacy 2 met een voldoende zijn afgerond. Het vak is gehaald bij een eindcijfer > 5,75 Indien er sprake is van onvoorziene omstandigheden, of maatregelen als gevolg van COVID-19, wordt als volgt afgeweken van voorgeschreven assessment: - het tentamen zal online worden afgenomen in de vorm van een open boek tentamen waar studenten drie uur de tijd voor hebben om het te maken. - de groepsopdracht blijft hetzelfde	

TB121B	Governance and Law 1	5
Module Manager	Mr. J.M. Kooijman	
Instructor	Dr.ir. E. Minkman	
Responsible for assignments	Mr. J.M. Kooijman	
Co-responsible for assignments	Dr. H.G. van der Voort	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	

TB122B	Micro and Market Economy	5
Responsible Instructor	S. Renes	
Module Manager	Dr. A.F. Correlje	
Instructor	Dr. A.F. Correlje	
Instructor	S. Renes	
Instructor	Dr.ir. R. van Bergem	
Responsible for assignments	Dr. A.F. Correlje	
Co-responsible for assignments	S. Renes	
Contact Hours / Week x/x/x/x	0/3/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

TB131B	Differential Equations and Linear Algebra	5
Module Manager	Dr. T.W.C. Vroegrijk	
Responsible for assignments	Dr. T.W.C. Vroegrijk	
Co-responsible for assignments	Drs. J.A. Verheij	
Contact Hours / Week x/x/x/x	5/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	

TB132B	Multivariable Analysis and Linear Algebra	5
Module Manager	Dr. T.W.C. Vroegrijk	
Responsible for assignments	Dr. T.W.C. Vroegrijk	
Co-responsible for assignments	Drs. J.A. Verheij	
Contact Hours / Week x/x/x/x	0/5/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

TB133D	Introduction to Programming with Python	5
Module Manager	Ir. I. Lefter	
Instructor	Ö. Okur	
Responsible for assignments	Ir. I. Lefter	
Co-responsible for assignments	Prof.dr. M.E. Warnier	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course provides an introduction to programming. The course covers the following parts: - Python and Jupyter Notebooks - Basic imperative programming concepts (expressions, statements, functions, conditionals, loops) - Basic data types (strings, lists, dictionaries, tuples, files) - Debugging - Object-oriented programming - Introduction to libraries (Numpys, Matplotlib)	
Study Goals	By the end this course, you should be able to: Implement Python programs using basic data types and programming concepts Apply Python programs using classes and objects Implement Python programs using algorithms Perform debugging to remove errors from complex Python programs	
Education Method	Mix of theory lectures and supervised exercises sessions. Programming assignments will be given which are used to practice the programming skills and will be worked on during the exercise sessions and in assignments. Reading material will be available.	
Literature and Study Materials	Book: Think Python How to Think Like a Computer Scientist - Allen Downey (Available online: https://greenteapress.com/thinkpython2/thinkpython2.pdf) Jupyter Notebooks with teaching material and practice assignments.	
Assessment	Computer exam	

TB134A	Statistics and Data Analysis	5
Module Manager	M. Kroesen	
Instructor	Dr. T.W.C. Vroegrijk	
Responsible for assignments	M. Kroesen	
Co-responsible for assignments	Dr. T.W.C. Vroegrijk	
Contact Hours / Week x/x/x/x	0/0/6/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

TB135C	Operational Research	5
Module Manager	Dr.ir. P.W. Heijnen	
Instructor	Dr.ir. P.W. Heijnen	
Instructor	Dr. P.W.G. Bots	
Responsible for assignments	Dr.ir. P.W. Heijnen	
Co-responsible for assignments	Dr. P.W.G. Bots	
Contact Hours / Week x/x/x/x	0/0/0/4,75	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Expected prior knowledge	--	
Parts	--	
Course Contents	--	
Study Goals	--	
Education Method	--	
Assessment	--	

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Technologiespecialisatie 1	
Director of Studies	Dr. J.A. Annema

TB141EB	Introduction to Energy and Industry Systems	5
Module Manager	Dr.ir. E.J.L. Chappin	
Instructor	Prof.dr.ir. M.P.C. Weijnen	
Responsible for assignments	Dr.ir. E.J.L. Chappin	
Co-responsible for assignments	Prof.dr.ir. M.P.C. Weijnen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	--	
Study Goals	--	
Education Method	--	
Literature and Study Materials	--	

TB141IC	ICT System Engineering and Rapid Prototyping	5
Module Manager	Prof.dr.ir. M.F.W.H.A. Janssen	
Instructor	Prof.dr.ir. M.F.W.H.A. Janssen	
Instructor	Dr.ir. J. De Stefani	
Co-responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	TB242IB - Interconnected engineering systems TB243IB - I&C Systems: Risk & Control TB341 Value Creation (from 2023/2024, due to I&C track redesign)	
Course Contents	The course introduces students to think and act as software engineer in TB context. The course integrated theory and practical work by developing a software prototype as final deliverable using the theory and practice behind software engineering, requirements formulating, and prototype assembling. Students are able to rapid prototype a functional software using low-code application. For this deliverable, the course presents and compare the most common software approaches, teaches how to develop diagrams using Unified Modelling Language (UML) diagrams, and, how to differentiate the most common programming paradigms. Finally the resulting prototype should be evaluated within the broader societal context.	
Study Goals	At the end of the course, the student should be able to: LO1: Describe the most common software design processes (at least Waterfall, Agile, Re-use based) LO2: Model real-life situations using UML diagrams (at least, use-cases, class and sequence diagrams) LO3: Explain the basic conceptual distinctions for characterizing IT systems, including the following: o LO3.a: Categorize programming languages in different levels of programming approaches (at least procedural, object-oriented, functional) o LO3.b: Define the basic components of computer hardware and hardware architectures LO4: Implement functional software using rapid prototyping approach and a low-code platform LO5: Identify the societal impact of the developed application.	
Education Method	Expected education methods (classroom): Lectures (presenting slides) UML practical work (making use-case, sequence and class diagrams) Software Workshops in format of Questions and Answers (Q&A) Low-code practical work (making a functional prototype) Elevator pitch students presenting low-code prototyping Expected education methods (digital COVID-19) Online lectures using Microsoft Teams (alternative Google Meets + Classroom + Youtube + Drive) Online workshops in format of Q&A Instructional videos recorded and stored at M$ Teams or Google Youtube	
Assessment	The course assessment is be composed of three parts: # Summative assignment: 30% of the final grade # Exam: - Part I - UML Modeling: 30% of the final grade - Part II - System engineering: 40% of the final grade If on campus activities are allowed, the assessment will take place as follows : - Summative Assignment : Submission via Brightspace + Student presentations on campus - Exam: On campus Computer Exam (via Mobius) In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed assessment will be modified as follows : - Summative Assignment : Submission via Brightspace + Student presentations remotely (via Zoom/Teams). - Exam: Remote computer Exam (via Mobius) Passing grade: To pass the course, the average grade of the three parts (Summative assignments + Exam Part I + Exam Part II) needs to be 5.8 or higher. Moreover, each part needs to have a grade of 5 or higher.	

TB141TB	Transport System Analysis	5
Module Manager	Dr. J.A. Annema	
Instructor	Prof.dr. G.P. van Wee	
Instructor	Dr. J.A. Annema	
Instructor	B. Pudane	
Responsible for assignments	Dr. J.A. Annema	
Co-responsible for assignments	Prof.dr. G.P. van Wee	
Contact Hours / Week x/x/x/x	0/0/5/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch English	
Course Contents	The objective of this course is to learn to understand and analyse transport systems. Transport systems have large positive impacts on our daily activities and quality of our lives as they ensure mobility, accessibility, and safety for people and companies and have enormous importance for the economy. At the same time, they are also associated with problems in the environment, traffic safety, and congestion. It is essential to view the current transport system in the context of both its positive and negative impacts on environmental, health, and safety. Policy makers and researchers often ask: how can we increase the accessibility that the transport system provides without exacerbating the problems of the system? Better still: how can we improve accessibility and at the same time solve congestion, environmental, safety, and health problems? In this course you will learn to analyse challenging policy questions such as the above. You will learn about the components of the transport system: technology (roads, cars, etc.), the behaviour of people and companies, and spatial developments. You will learn how these elements depend on each other (using conceptual models), and how together they lead to positive and negative transport system effects. Finally, you will learn computation techniques and models that you can use to quantitatively estimate the impacts of any transport policy on environment (in particular, air quality), traffic safety, accessibility, and public health.	
Study Goals	Upon the successful completion of this course you will be able to: 1. identify the different elements of the transport system. 2. describe the relationship of the different elements in the transport system. 3. create conceptual models of the transport system and use them to analyse transport policies and technological scenarios. 4. compute quantitative effects of transport policies and technological scenarios following provided calculation models.	
Education Method	Students independently study the textbook "The Transport System and Transport Policy" (van Wee et al., 2013), as well as additional materials and videos provided by the teachers. During lectures, the students will work in groups on creating conceptual models of transport system impacts and on quantitative exercises. Groups are also expected to meet outside of the class to prepare a presentation and written assignment, and to complete the quantitative exercises. The group work is designed for the students to learn to analyse the transport system elements and their interrelationships, and to estimate effects quantitatively based on those relationships. Each group works independently while receiving support from the course instructors.	
Assessment	Learning objectives 1 to 4 are tested with an individual exam (60% of the final grade) and learning objectives 2 to 4 also with a group assignment (40%). In order to successfully complete this course, the weighted average grade must be at least 5.75, as well as the two components (exam and group assignment) must each be at least 5.0. In the event of unforeseen circumstances or COVID19 measures, the individual exam will be an open book online exam lasting 3 hours.	

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Technologiespecialisatie 2	
Director of Studies	Dr. J.A. Annema

TB142EA	Analysis of Energy Systems	5
Module Manager	Dr.ir. L. Stougie	
Responsible for assignments	Dr.ir. L. Stougie	
Co-responsible for assignments	Dr.ir. R.M. Stikkelman	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	

TB142IC	Data Science in Technische Bestuurskunde	5
Module Manager	Prof.dr.ir. M.F.W.H.A. Janssen	
Instructor	R.I.J. Dobbe	
Co-responsible for assignments	Dr.ir. G.A. de Reuver	
Contact Hours / Week x/x/x/x	0/0/0/14 - 6 in class, 8 homework	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Course Contents	Today's information, communication and computing technology makes it possible for organizations to monitor their activities at almost every level, with the result that large amounts of data are collected, which in turn can form an input for other processes and applications. Processing collected data into information that is usable, reliable and secure is a delicate process. As a technical public administrator you must be able to make the translation between the administrative process and the technically available tools and methods. It requires an integrated look at the context of application as well as the life cycle of data, from collection to processing in analyzes and AI systems. In addition, it is essential to understand how AI systems behave and process data, and what ethical, legal and social implications these can have.	
Study Goals	LO1: Sociotechnical specification - Identify how AI systems may contribute to the formulation and solution of multi-stakeholder problems, by: a) determining and operationalizing a question such that an intelligent system or data analysis could produce a meaningful knowledge. b) exploring a data set to determine whether and how it might inform questions of interest or aid decision-making c) presenting the affordances and limitations of an AI system with appropriate visuals and written argumentation d) providing recommendations for how the AI system may be used in practice LO2: Design - Execute the design of an AI system for decision-making, including: a) selecting the appropriate modeling approaches and algorithms b) determining the need and quality of data inputs c) processing data sets using modern data science scripts and statistical tools d) building an AI model (rule-based or learning-based) for decision-making e) summarizing key mathematical concepts for AI systems f) apply relevant tests to assess a decision-making system for various metrics g) identifying the limitations of the AI system outputs and assessing its impact on correctness and utility LO3: Reflection - Communicating the affordances and limitations of a data-driven and algorithmic decision-making system in a responsible manner, including: a) using best practices for documenting the analyses for reproducibility and accountability b) reflecting on the ethical, legal and societal implications of an AI system in terms of the research process and implementation c) translating implications to possible tradeoffs in the problem formulation and use of the system	
Education Method	Main activities: - In-class instruction and discussion - Practical assignments - Group project - Reflection assignments	
Assessment	Practical lab assignments - 2/10 of grade A final reflection essay - 2/10 of grade A group project - 3/10 of grade A final exam - 3/10 of grade	

TB142TA	Logistics 1	5
Module Manager	Dr. J.H.R. van Duin	
Responsible for assignments	Dr. J.H.R. van Duin	
Co-responsible for assignments	Dr. S. Fazi	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Remarks	The lectures are partly taught in English and Dutch due to the composition of the teachers' team.	

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2e jaar BSc TB 2022	
Director of Studies	Dr. J.A. Annema

TB211A	Analysis of Multi-Actor Systems	5
Module Manager	Dr.ir. C. van Daalen	
Instructor	Dr.ir. M.P.M. Ruijgh-van der Ploeg	
Instructor	Dr. L.J. Kortmann	
Instructor	Dr.ir. V.J. Cortes Arevalo	
Instructor	Dr. M. Freese	
Co-responsible for assignments	Dr. P.W.G. Bots	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Tags	Group work Intensive Policy Analysis Project	

TB212B	Ethics and Safety	5
Module Manager	Dr. S.M. Copeland	
Instructor	K.L.L. van Nunen	
Co-responsible for assignments	Prof.dr.ir. P.H.A.J.M. van Gelder	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch English	

TB221B	Economy of Infrastructures	5
Module Manager	Dr. A.F. Correlje	
Instructor	Prof.dr. R. Künneke	
Instructor	Dr. A.F. Correlje	
Instructor	Dr.ir. R. van Bergem	
Instructor	S. Renes	
Responsible for assignments	Dr. A.F. Correlje	
Co-responsible for assignments	Prof.dr. R. Künneke	
Contact Hours / Week x/x/x/x	2/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	

TB222A	Governance and Law 2	5
Module Manager	Mr. J.M. Kooijman	
Responsible for assignments	Mr. J.M. Kooijman	
Co-responsible for assignments	Dr. H.G. van der Voort	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

TB223C	Organisation and Management	5
Module Manager	Dr. M.L.C. de Bruijne	
Instructor	Prof.mr.dr. J.A. de Bruijn	
Instructor	Dr. H.G. van der Voort	
Instructor	Dr.ir. G. Bekebrede	
Instructor	Dr. G. de Vries	
Responsible for assignments	Dr. W.W. Veeneman	
Co-responsible for assignments	Dr.ir. G. Bekebrede	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	
Course Contents	See Dutch (course is in Dutch)	
Study Goals	See Dutch (course is in Dutch)	
Education Method	See Dutch (course is in Dutch)	
Assessment	See Dutch (course is in Dutch)	

TB231C	System modelling 3: simulation methods	5
Module Manager	Dr.ir. W.L. Auping	
Instructor	Dr.ir. I. Nikolic	
Instructor	Dr.ir. W.L. Auping	
Responsible for assignments	Dr.ir. I. Nikolic	
Responsible for assignments	Dr.ir. W.L. Auping	
Co-responsible for assignments	Dr. O. Kammouh	
Co-responsible for assignments	F.M. d'Hont	
Contact Hours / Week x/x/x/x	6/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	
Required for	This course is required for students in the Technische bestuurskunde bachelor programme, and is a prerequisite for the project in TB233B Systems Modelling 4 simulation methods project.	
Expected prior knowledge	It is expected from students that they have basic knowledge about systems modelling, differential equations and linear algebra, and basic programming skills in Python.	
Parts	This course consists of two parts: - Agent-Based Modelling - System Dynamics	
Course Contents	In System Modelling 3: Simulation methods two different simulation methods will be taught: Agent-Based Modelling (ABM), and System Dynamics (SD). There will be a lot of emphasis on practising conceptualisation, formulation, evaluation, and use of models, in which the theory is aimed at head-starting students, and to better understand different concepts and background of these two modelling methods.	
Study Goals	Upon completion of this course, students should be able to - explain the process of modelling for supporting decision-making on large societal challenges; - conceptualise, formulate, parameterise, test, and use a model of a complex, societal policy problem using SD and ABM formalisms; - analyse a complex societal policy problem using an SD model and ABM; - identify and test interventions for addressing the policy problem using the developed simulation model; - assess and argue whether SD and/or ABM are appropriate for analysing a policy problem; - link system structure to possible dynamic behaviour of important indicators of a policy problem. For SD, this system structured is aggregated. For ABM, this system structure is microscopic. SD specific - develop and use a medium complexity SD model in Vensim ABM specific - develop and use a medium complexity ABM model in MESA.	
Education Method	This course uses two main methods of education: - Lectures for the theoretical aspect of the course - Lab sessions to develop skills for working with the tools Vensim and MESA	
Computer Use	Agent-Based Modelling: The ABM simulation models will be made with MESA. MESA is a Python 3 based framework for ABM. Student will need to have a working Python 3 installment before they install MESA. System Dynamics: The SD simulation models will be made with Vensim. Students can request a temporary Vensim license for Windows or iOS. This program is also installed in the computer rooms in the Faculty (TPM in Delft) and can be used via TU Delft's weblogin.	
Course Relations	This course expands on the knowledge obtained in TB131B Differential Equations and Linear Algebra, TB112C Systems modelling 1: modelleerprincipes, TB133D Introduction to Programming with Python, and TB113B Systems Modelling 2: Complex Systems. Especially the Python skills obtained during Systems Modelling 2 are indispensable during this course.	
Literature and Study Materials	Agent-Based Modelling: Jupyter notebooks and lecture handouts System Dynamics: E-book for theory and exercises	
Assessment	This course is examined in a written, online, open-book computer exam. If the Corona measures allow, the exam will be on campus. with two separate parts: - Agent-Based Modelling (50%) - System Dynamics (50%) The final grade is composed of partial grades for each part (weights as indicated on Osiris). Partial grades count when they are at least a 5.0 (compliant with article 14, 6b of the Rules and Guidelines of the Board of Examiners of the TB bachelor programme).	
Targetgroup	Students from the Technische Bestuurskunde bachelor programme.	

TB232B	Research and Data-analysis	5
Module Manager	M. Kroesen	
Instructor	Dr. E.J.E. Molin	
Instructor	Ir. S. van Cranenburgh	
Responsible for assignments	M. Kroesen	
Co-responsible for assignments	Dr. E.J.E. Molin	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	

TB233B	System modelling 4: project simulation methods	5
Module Manager	Dr. C.N. van der Wal	
Instructor	Dr.ir. W.L. Auping	
Instructor	M.B. van der Meulen	
Co-responsible for assignments	Dr.ir. C. van Daalen	
Contact Hours / Week x/x/x/x	0/0/2/0	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	The prior knowledge for this course is developed during the following courses: - TB112C System's modelling 1: modelling principles; - TB113B System's modelling 2: complex systems; - TB231C System's modelling 3: simulation methods. Het last course, System's modelling 3, is the most important of these 3 courses.	
Course Contents	This project course will be primarily executed by groups containing 2 students. Each group will choose a modelling method: either system dynamics (SD), or agent-based modelling (ABM), and a socio-technical problem that they will model quantitatively. The socio-technical problem that will be selected will be designed such that it can be modelled both with SD and ABM. Each group will independently, but under the supervision of teaching assistants and the course lecturers, apply the different phases of the modelling cycle. Next, each group will use their own model to analyse a research problem and report this in a scientific article. In parallel to the modelling and simulation process, the academic writing skills of the students will be developed by the use of writing exercises and workshops. The skill level in academic writing will be assessed explicitly. Each group will receive the task to write peer reviews for a group that used the other modelling paradigm. In week 6, this will concern the model and the model description. In week 8, the results of both different methods can be compared, and the scientific article of the other group will be assessed on contents and writing skills.	
Assessment	The grade will be composed in the following way: - 20% academic writing - 20% model quality - 60% reporting quality - The two peer-reviews will count for the final grade and can, depending on their quality, result in a bonus or subtraction of the final grade within the range: [-0.5, +0.5]	

TB234B	Multivariate Data Analysis	5
Module Manager	Dr. E.J.E. Molin	
Responsible for assignments	Dr. E.J.E. Molin	
Co-responsible for assignments	M. Kroesen	
Contact Hours / Week x/x/x/x	0/6/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

Year	**2022/2023**
Organization	**Technology, Policy and Management**
Education	**Bachelor Systems Engineering, Policy Analysis & M**

Technologiespecialisatie 3	
Director of Studies	Dr. J.A. Annema

TB241EA	Transport Phenomena	5
Module Manager	Dr.ir. I. Bouwmans	
Module Manager	Dr.ir. G. Korevaar	
Module Manager	Prof.dr.ir. C.A. Ramirez Ramirez	
Responsible for assignments	Dr.ir. I. Bouwmans	
Responsible for assignments	Dr.ir. G. Korevaar	
Co-responsible for assignments	Prof.dr.ir. C.A. Ramirez Ramirez	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	Dutch	

TB241IB	Object Oriented Programming	5
Module Manager	Dr. N. Metoui	
Instructor	S.H. van Engelenburg	
Instructor	Dr. N. Metoui	
Instructor	Dr.ir. J. De Stefani	
Co-responsible for assignments	A. Ghorbani	
Contact Hours / Week x/x/x/x	4/0/0/0	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This course requires: - Basic knowledge of UML Diagrams. - Basic knowledge of any (scripting or programming) language.	
Course Contents	This course deepens your skills and conceptual understanding of programming. You will learn object- oriented Design (OOD) with UML and Object-Oriented Programming (OOP) with Java. UML and Java are industry standard design and programming languages used to develop enterprise software, desktop apps, web apps, mobile apps, games and much more. In this course, you will learn how to position the languages you learned so far, and what to use when: Low-code (TB141Ic), scripting (TB133d; TB142Ic) and object-oriented (TB241Ib). You will also familiarize yourself with Java development tools and practices that allow you to write clean and reusable code and to understand and modify existing ones easily. The skill set you will earn in this course can be useful, in the future, to design and build software tools and components, to update or upscale software, or to manage a java project.	
Study Goals	At the end of this course, you will be able to: LO1 - Identify key characteristics of the Java programming language and development environment (EDI, SDK, JVM, etc.) LO2 - Use basic Java APIs and components (data types, data structures, algorithmic structures, and methods) LO3 - Apply key elements, concepts, and design patterns of object-oriented programming LO4 Write clean, understandable, and reusable code following Java coding best practices LO5 - Develop object-oriented programs using Java.	
Education Method	In this course, you will have alternating lectures and hands-on labs	
Literature and Study Materials	Materials (slides, book chapters, tutorials, references to software packages, etc.) will be made available via brightspace.	
Assessment	70% Individual tests and assignments 30% Group project In case of unforeseen circumstances or measures resulting from COVID-19, tests and assignments will be done (timed) online.	

TB241TB	Logistics 2	5
Module Manager	Dr. S. Fazi	
Co-responsible for assignments	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	4/0/0/0 excl 10 hours of self study	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	According to the Council of Supply Chain Management Professionals logistics is defined as the process of planning, implementing, and controlling procedures for the efficient and effective transportation and storage of goods including services, and related information from the point of origin to the point of consumption for the purpose of conforming to customer requirements. Logistics activities should be in line with other business activities such as marketing and purchasing, in order to create sustainable competitive advantages for individual companies and service providers, and for the supply chain as a whole. Given the strategic importance of logistics for individual organizations and supply chains, it is vital to know how the performance of logistics activities is measured, and how these activities can be modelled and improved. Therefore in this course different logistics aspects are discussed and analyzed. A selection of quantitative methodologies are introduced to model and provide solutions to a wide range of relevant problems in logistics. We focus on two classes of logistics problems: Logistics problems within organizations: forecasting and demand modeling; inventory management; clustering and classification of items. Distribution problems. Methodologies include analytical models, linear programming models, basic data mining techniques.	
Study Goals	At the end of this course you will be able to: Analyze relevant aspects of logistics and transportation system: inventory, routing, location, data; Formulate (and solve) a selection of relevant logistics problems; Select and apply an appropriate methodology for a given logistics problem; Find possibilities to improve/adapt a logistics system through data and analytical models; Model operational aspects through mathematical linear programming models; Understand the integrative role of logistics function in and between organizations;	
Education Method	Lectures Assignments Self-study It is expected that for academic year 2022/2023 the educational activities will be held on-campus. Consultation hours will be held both online and on-campus.	
Literature and Study Materials	A selection of chapters from: 1) Reid, R. D., Sanders, N. R., Operations Management: An Integrated Approach, 5th edition, Wiley, 2013. 2) Daganzo, C.F., Logistics Systems Analysis, 4th edition, Springer-Verlag, 2010. 3) Zaki, Mohammed J., Wagner Meira Jr, and Wagner Meira. Data mining and analysis: fundamental concepts and algorithms. Cambridge University Press, 2014. 4) A booklet provided at the start of the course 5) Slides and exercises	
Prerequisites	Logistiek 1, Basic mathematics	
Assessment	The course consists of two parts with the following weighting: Logistics problems within organizations: 50%. Distribution problems: 50%. Grading: One written exam: 80% Assignments: 20% The passing grade is 5.8, calculated as: 0.8*(final exam) + 0.2*((Assignment 1 + Assignment 2)/2). Minimum grade for final exam: minimum grade is 5.0 Minimum grade for assignments: average grade at least 5 -> (Assignment 1 + Assignment 2)/2 >= 5.0 In case of unforeseen circumstances or measures resulting from COVID-19, the final exam will take place in an online format: a take-home exam with a limited duration. The student will receive the exam via email at the start of the exam session and must	

	upload the answers in Brightspace within the time limit. In case the regular on-campus exam can be held, no online possibility will be offered.
Exam Hours	3 hours

Year	**2022/2023**
Organization	**Technology, Policy and Management**
Education	**Bachelor Systems Engineering, Policy Analysis & M**

Technologiespecialisatie 4	
Director of Studies	Dr. J.A. Annema

TB242EB	Processes in the Energy Sector	5
Module Manager	Dr.ir. L. Stougie	
Instructor	Dr.ir. L. Stougie	
Responsible for assignments	Dr.ir. L. Stougie	
Co-responsible for assignments	Dr.ir. I. Bouwmans	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

TB242IB	Interconnected Engineering Systems	5
Module Manager	Dr. Y. Ding	
Instructor	Prof.dr.ir. M.F.W.H.A. Janssen	
Instructor	R.I.J. Dobbe	
Instructor	Dr. Y. Ding	
Responsible for assignments	Dr. Y. Ding	
Co-responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Course Contents	Internet is a global system that has interconnected various kinds of systems and services worldwide. This course analyzes the architecture and core technologies of the Internet from systemic and interdisciplinary perspectives. As a hands-on oriented course, we learn network programming (using python) and apply the learned programming skills in practical settings. Attention is paid to the increasing importance of interconnected environments, in particular the networking of engineering systems and services through digital platforms. We analyse the concept of interconnected engineering systems on the basis of concrete cases, e.g., connected cars, cyber-physical systems, as means to illustrate the complex interactions of technology and society. In addition, a number of modern development and research themes will be discussed for the evolution of distributed information and communication systems, e.g., new networking and computing techniques for large-scale deployment of the Internet of Things (IoT). On top of the key technologies, we consider the social context to which the high expectations of the future Internet technologies need to tackle in terms of security, privacy risks.	
Study Goals	After the course, students are able to: - Illustrate Internet architecture, design principles and routing algorithms - Evaluate computing paradigms and wireless & mobile communication in practical scenarios - Design networking programming solutions for practical problems - Apply Internet of Things (IoT) and cyber-physical systems (CPS) technologies in case studies - Analyze security & privacy risks, challenges and solutions for interconnected engineering systems such as IoT	
Education Method	Lectures, guest lectures, individual and group work, programming tutorials and practice	
Assessment	- formative mandatory assignments - summative final exam In case of unforeseen circumstances or measures resulting from COVID-19, the prescribed summative assessment will be conducted as time-based take-home exam.	

TB242TB	Improving the Transport System	5
Module Manager	Dr. J.A. Annema	
Instructor	Dr. J.A. Annema	
Instructor	N. Mouter	
Responsible for assignments	Dr. J.A. Annema	
Co-responsible for assignments	N. Mouter	
Contact Hours / Week x/x/x/x	0/x/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	Dutch	

Year	**2022/2023**
Organization	**Technology, Policy and Management**
Education	**Bachelor Systems Engineering, Policy Analysis & M**

Technologiespecialisatie 5	
Director of Studies	Dr. J.A. Annema

CTB1420-17	Transport en Planning	5
Responsible Instructor	Dr.ir. A. Gavriilidou	
Instructor	Dr. V.L. Knoop	
Practical Coordinator	Dr.ir. A. Gavriilidou	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	
Collegerama	No	

TB243EA	Analysis of Industrial Systems	5
Course Coordinator	Dr.ir. P. Ibarra Gonzalez	
Module Manager	Prof.dr.ir. Z. Lukszo	
Instructor	Dr.ir. P. Ibarra Gonzalez	
Responsible for assignments	Prof.dr.ir. Z. Lukszo	
Co-responsible for assignments	Dr.ir. L. Stougie	
Contact Hours / Week x/x/x/x	0/0/0/6	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Expected prior knowledge	TB141E, TB142E, TB241E	
Summary	The course aims to teach students a number of relevant production processes from the level of unit operations up to the system level. The students will learn how to recognize the basic elements of a production process (flowsheet-graphical way of describing a process) and understand their role and sequence. It also gives an overview of important industries in the Netherlands.	
Course Contents	The main topics covered in this course are: - Structuring industrial production processes - Separation and reaction processes - Wastewater treatment and drinking water preparation - Batch and Continuous Processes - The methanol process as an example of a continuous process - The brewery as an example of a batch process - The crude oil system: Refinery - Petrochemical industry - Overview of Processes in the Agro-Industry - Overview of Processes from the Biobased Industry - Industrial processes in the Netherlands	
Study Goals	By the end of the course, you should be able to: LO1- Identify and name the feedstock, unit operations, and products of selected industrial production processes LO2- Design industrial processes using flow sheeting tools LO3- Describe the processing steps, required unit operations, and product chains of selected industrial production processes LO4- Explain the basic functioning and types of reactors and separation units in selected industrial production processes LO5- Calculate the performance of a simple production process through mass and energy balances LO6- Calculate product conversion for different types of reactors LO7- Determine variables influencing the performance of separation processes	
Education Method	Prior reading of course material (self-study), interactive lectures, instruction lectures (tutorial-style), class assignments, and homework assignments.	
Computer Use	Use of a laptop is needed to answer the assignments in Excel	
Literature and Study Materials	If required, before each lecture extra material for reading will be provided.	
Books	A course manual will be provided	
Assessment	Formative assessments during lectures, both individually and in groups (oral participation, and short multiple-choice tests). The summative assessment is in the form of a final exam in week 10 and has a duration of 3 hours. The exam consists of 2 parts. The first part consists of multiple-choice questions that count for 50% of the final course mark. The second part includes open questions that count for the other 50% of the final course mark.	
Exam Hours	3	

TB243IB	I&C Systems: Risk & Control	5
Module Manager	Dr. Y. Ding	
Instructor	S.H. van Engelenburg	
Co-responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	Dutch	
Course Contents	This module is about assessing and overcoming risks in ICT to ensure systems are maintainable in the long term in complex multi-actor environments and that they do not create new risks. You learn to understand the different hardware components that make up ICT systems and the different organisational, inter-organisational and large-scale systems designs. You learn how to use this knowledge to assess the risks different ICT systems pose for individuals, organisations and society. You will also learn the different strategies to reduce these risks and determine if and how to apply them. The theoretical background will be explained during the course during different lectures and guest lectures. In different groups, you will then apply the theory to several cases in which you will assess risks and identify strategies for reducing them. You will present your case to the other students in the course, after which they can ask questions about it, and there will be a more in-depth discussion of the case, the strategies chosen, and the considerations made.	
Study Goals	- To describe the hardware components of ICT systems (network devices, servers, workstations). - To describe the different designs for organisational, inter-organisational and large-scale systems. - To describe different hardware and management strategies for reducing risks in ICT. - To recommend an appropriate strategy for reducing risks for an ICT system in a given case. - To describe the risks of ICT systems on the systems (hardware) level, within organisations and on an inter-organisational and societal level. - To assess the risks of an ICT system on the systems (hardware) level, within organisations and on an inter-organisational and societal level in a given case.	
Education Method	Lectures, assignments, self-study	
Assessment	The assessment during the course consists of two parts. The first one is an individual exam, which contains a mix of knowledge questions and simple cases for which you will need to assess and address risks. The individual exam will be 60% of your final grade. The second part of your assessment will be based on a final case assignment for which you will assess risks and identify a strategy for dealing with that risk in your group. For this part, the assignment report and its presentation will be 40% of your final grade. You need to score at least a 5 for each part and at least a 5.75 for both parts together to pass the course	

Year	**2022/2023**
Organization	**Technology, Policy and Management**
Education	**Bachelor Systems Engineering, Policy Analysis & M**

3e jaar BSc TB 2022

Year	**2022/2023**
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Education	**Bachelor Systems Engineering, Policy Analysis & M**

Verplichte vakken 3e jaar 2022	
Director of Studies	Dr. J.A. Annema

TB323A	Governance van sociotechnische systemen	5
Module Manager	Dr. R.S. van Wegberg	
Instructor	Dr. H.G. van der Voort	
Instructor	Dr. M.L.C. de Bruijne	
Instructor	Dr.ir. E. Minkman	
Instructor	Dr. R.S. van Wegberg	
Co-responsible for assignments	Dr. M.L.C. de Bruijne	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	Dutch	

TB351D	Bachelor Final Project	15
Module Manager	M. Kroesen	
Co-responsible for assignments	Dr. P.W.G. Bots	
Contact Hours / Week x/x/x/x	2/2/2/2	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	Dutch	

Governance specialisatie (E&I / I&C / T&L)	
Director of Studies	Dr. J.A. Annema

TB321EB	Governance specialisation (E and I)	5
Module Manager	Prof.dr.ir. L.J. de Vries	
Instructor	M. Looij	
Co-responsible for assignments	Dr. A.F. Correlje	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

TB321IB	Governance specialisation (I and C)	5
Module Manager	A.M.G. Zuiderwijk-van Eijk	
Instructor	M. Looij	
Co-responsible for assignments	Dr. J. Ubacht	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

TB321TB	Governance specialisation (T and L)	5
Module Manager	Dr. J.A. Annema	
Instructor	M. Looij	
Co-responsible for assignments	N. Mouter	
Contact Hours / Week x/x/x/x	0/0/x/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	Dutch	

Year	**2022/2023**
Organization	**Technology, Policy and Management**
Education	**Bachelor Systems Engineering, Policy Analysis & M**

Technologiespecialisatie 6	
Director of Studies	Dr. J.A. Annema

TB341EC	Performance analysis in Energy and Industry	5
Module Manager	Dr.ir. P.W. Heijnen	
Instructor	Dr. P.W.G. Bots	
Co-responsible for assignments	Dr. P.W.G. Bots	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3A 3B 4A	
Course Language	Dutch	

TB341IC	I and C Risk and Control	5
Module Manager	Dr. Y. Ding	
Instructor	S.H. van Engelenburg	
Co-responsible for assignments	Prof.dr.ir. M.F.W.H.A. Janssen	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Expected prior knowledge	TB243IA, TB142IB	
Course Contents	This module is about operating information infrastructures, including aspects of data security and privacy for ICT innovations. You learn to understand, analyze and specify the relationships between different information systems, the processes and organizational factors to approach risk management, as well as foundation of computer security. This course mainly concerns organizational and structural approaches to prevent security risks from materializing as well as necessary background on these security risks. The main idea is: We know how to securely run systems. We just have to do this. In the first weeks of the module, basic concepts are provided that are tested in the interim examination halfway through the module (e.g. in week 5). Thereafter, the technical content on how to manage, and take over mismanaged, IT systems is deepened in the second half of the course. This second part will be tested with a final exam.	
Study Goals	By the end of this course, students will be able to: - Reproduce concepts, methods, and tools from IT security and ICT risk management. - Analyze the risks involved in concrete case-studies of ICT systems and use-case. - Be able to improve the concrete case-studies by appropriately selecting concepts, methods, and tools from IT security and ICT risk management - Report their analysis and proposal in a written document - Take a managerial role in operating an IT system, collaborating with technical employees in their organizations	
Education Method	Lectures, work groups, self-study	
Literature and Study Materials	This course is based on 'The practice of system and network engineering' Vol. 1 The Practice of System and Network Administration Volume 1: DevOps and other Best Practices for Enterprise IT ISBN 9780321919168 This book is available as an eBook via the TU Delft library. Students are expected to read the chapters relevant for each lecture before the lecture. Reading for all lectures will be announced at the beginning of the course.	
Assessment	This course has a mid-term and final exam. Both must be at least graded with 5.0. The final grade consists out of the average (50/50) of the two course parts, and the rounded grade must be at least a 6.0 (5.75 rounded up). Questions in the exam will expand on material from the course and include real-world examples. The exam will be closed-book. For both exams there will be a dedicated exam preparation lecture in the week of the exam. If in-person examination is not possible, the exam will become a take-home exam, submitted via brightspace. While that exam will be designed to take around 3 hours, students will receive a timeslot of at least 24h during which they can work on the assignment following their own scheduling.	
Enrolment / Application	Enroll in the Brightspace module, where all required materials and information can be found	

TB341TC	Quantitative Models for Transport	5
Module Manager	Dr. S. Fazi	
Instructor	B. Pudane	
Instructor	Dr. S. Fazi	
Co-responsible for assignments	Dr. J.A. Annema	
Contact Hours / Week x/x/x/x	0/0/4/0	
Education Period	3	
Start Education	3	
Exam Period	3 4A	
Course Language	English	
Course Contents	Many transportation problems are tackled with quantitative methods, due to the necessity for the decision-makers to quantify relevant aspects: service frequencies, sequence of visits, costs, timings, etc. In this course, we will propose three fundamental methodologies in the transport domain to give answers to problems related to passengers mobility and freight transport. In the first part of the course, Discrete choice modelling will be proposed to solve passengers mobility related problems. The goal is to understand how individuals choose between different transport options based on a set of features and related utility functions. In the second part, simulation and mixed-integer linear programming models will be proposed to model both operational and strategic decisions. The proposed models are extensively used in container transport, parcel delivery, and trucking problems. Commercial optimization and simulation software will be used to solve the models and analyse the solutions.	
Study Goals	At the end of this course you will be able to: - Apply basic discrete choice modelling techniques - Perform statistical analysis to draw the main elements of the choice models - Model transportation systems by means of conceptual models and simulation - Recognize and develop the right mathematical model for a given transportation problem; - Analyse solutions and perform sensitivity analyses with ad-hoc software.	
Education Method	Lectures Assignments Self-study It is expected that for academic year 2022/2023 all the educational activities will be held on-campus. Consultation hours will be held both online and on-campus.	
Assessment	The course consists of two parts with the following weighting: Discrete choice models: 50%. Combinatorial linear mathematical models and simulation: 50%. Grading: One written exam: 80% Assignments: 20% The passing grade is 5.8, calculated as: 0.8*(final exam) + 0.2*((Assignment 1 + Assignment 2)/2). Minimum grade for final exam: minimum grade is 5.0 Minimum grade for assignments: average grade at least 5 -> (Assignment 1 + Assignment 2)/2 >= 5.0	

Keuze minor (zie minor lijst) 2021

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